Sec 7.3 (Separable DE's)

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We so far have looked at what a differential equation is and what a solution to a DE is and solved very simple equations like y'=2x -> y=x^2 +c But there are differential equations that we can not just directly integrate.

Ex. $\frac{dy}{dx} = 4x^2y^3$

How can we solve this equation. Well this equation is what we call a separable differential equation and we can solve it with the method of separation of variables.

Definition: Separable differential equation. A separable differential equation is one in which we can write our derivative as a product or quotient of two functions each only depending on one variable.

Ex. $\frac{dy}{dx} = g(x)f(y)$ or $\frac{dy}{dx} = \frac{g(x)}{f(y)}$ or $\frac{dy}{dx} = \frac{f(y)}{g(x)}$

$Ex\ \frac{dy}{dx} = 4x^2y^3$

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Method of Separation of variables.

To solve a seperable equation like $\frac{dy}{dx} = g(x)f(y)$ we write it in differential form

$$\frac{1}{f(x)}dy = g(x)dx$$

That is to say I separate all of the x stuff and y stuff on either side of the equation. I can then integrate both sides.

Ex. Solve $\frac{dy}{dx} = 4x^2y^3$

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Separate and solve the following DE's With the given initial condition.

$\frac{dy}{dx} = \frac{\ln(x)}{xy}$ $y(1) = 2$

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$\frac{du}{dt} = (2t + \sec^2 t)\frac{1}{2u}$ u(0)=-5

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$$xln(x) = y\left(1 + \sqrt{3 + y^2}\right)y'$$